

Chapter 4 Review

Learning Objectives

- Review all concepts from Data Representation and Processing
- Demonstrate binary conversion and data representation skills
- Apply chapter knowledge to solve a real-world problem

Engage (5-7 minutes)

Think of a harvest festival where farmers display all their crops together. Today we display everything we learned about data—number systems, text, images, audio, processing, storage, compression, and encryption.

Explore (10-12 minutes)

Complete a revision worksheet covering binary conversion, ASCII values, pixel calculations, audio sampling, compression types, and encryption concepts. Work in pairs and help each other with difficult questions.

Explain (10-12 minutes)

Chapter 4 covered data fundamentals, number systems, binary conversion, how text images and audio are represented digitally, data processing with IPO cycle, storage devices, compression techniques, and encryption for security. All concepts connect—binary underlies everything.

Elaborate (8-10 minutes)

Scenario challenge: A farmer wants to store 500 photos, 100 songs, and text records on a device. Calculate total storage needed, recommend a device, suggest compression for photos, and explain how encryption protects his market price data. Write your complete plan.

Evaluate (5-8 minutes)

Self-assessment: Rate your understanding of each topic on a scale of 1-5: binary, text encoding, image representation, audio representation, data processing, storage, compression, encryption. Identify your strongest and weakest areas.